Questioning Dominant Narratives

Overview

28 Jan 2026

- **question all assumptions of "big data" and "data science" including:**
  1. **scalability**
  2. **inclusion**
  3. **objectivity**
  4. **limitations and restrictions**
- **spark conversations around these assumptions**

Readings

On the Dangers of Stochastic Parrots: https://doi.org/10.1145/3442188.3445922
Critical Questions for Big Data: https://doi.org/10.1080/1369118X.2012.678878
Against Scale: http://arxiv.org/abs/2010.08850
Terms of Inclusion: http://arxiv.org/abs/2010.08850
Excavating Awareness and Power in Data Science:
https://doi.org/10.1177/20539517211040759

On the Dangers of Stochastic Parrots

Challenges in the rapid scaling of LLMs	• environmental **impact**
	  ○ **massive** computing **resources**
	  ○ **increased** carbon **emissions**
	• **bias** reproduction
	  ○ **training data** often contains **harmful** societal **bias**
	• **illusion** of **understanding**
	  ○ models **mimic** language **without** real **comprehension**
	• **incentive** structures
	  ○ industry pressures **prioritize larger** models **over ethics**

Key Concerns	• **larger isn't** always **better** • raises **questions** about **accountability**, **transparency**, and **who benefits** from AI systems • sparks **interests** over **ethics**, **governance**, and research **freedom**
Analysis	• this paper serves as a **critique** of **AI development** practices and culture and **urges** readers to think **ethically**, and **politically** about system **design** and **deployment**

"Critical Questions for Big Data" — Information, Communication & Society

Key Concepts	• **big data** is **more** than **big datasets**

	• it is a **socio-economic** phenomenon involving **technology**, **analysis**, and a **belief** system

Supporting Arguments	• **big data** is often **framed** as **neutral** and **objective** which is **misleading**
	• **questions** raised:
	  ○ can **big data improve public** goods or **entrench surveillance**?
	  ○ does BD broaden research, or is it limiting our focus

Analysis	• this **article pushes** readers to see **BD** as **embedded** in **power** relations and frames **analytics** as both a **scientific** and **social motivator**
Terms of Inclusion: Data, Discourse, Violence — New Media & Society	
Key Concepts	• **inclusion rhetoric** in data **ethics** can **mask** deeper **structural violence** • **inclusion without power shifts** can **normalize inequality** rather than challenge it
Supporting Arguments	• **inclusion** is **erroneously framed** as a **solution** for **bias** ○ this can **diffuse** radical **change** • **inclusion** efforts **risk co-optation** by existing power structures **if** they are **not challenged**
Analysis	• this article **challenges assumptions** about **inclusion** as a **goal** • even **inclusion efforts** have a **limiting agenda** • **shifts discussion** from surface **reforms** to structural **critique**
Excavating Awareness and Power in Data Science — Big Data & Society	
Key Concepts	• researchers must **practice 'trustworthiness'** in pervasive data **research** • especially when that research leaves **digital traces** of **indiviuals** • **awareness**, **power**, and **reflexivity** build **ethical** research approaches by **drawing** from '**ethnographic practices'**
Supporting Arguments	• **participant unawareness** in data research is a major **ethical issue** • **awareness** and **power relations** should be at the **foreground** in **methodology**

Analysis	• this paper defines '**epistemic relations'** – **who knows** what, who **consents** – by engaging in **ethical discourse** on **outcomes**, **bias**, and **fairness**
Async Materials	
A Common Lexicon	• **big data** is **less** about the **size**, and **more** about **capacity** to **search**, **aggregate**, and **cross-reference** large data sets • **data sets** that **were** once only **available** to **professional** data science, are **now widely available** • data **centers** are **factories** that **contribute** to **pollution** • **technological solutionism** : **technology can solve all of our problems** • can our **reliance** on **apps harm** our critical **thinking** or social **skills**?
Big Data - Big Challenges	• big **data** can **create** big **solutions** • **twitter** opening its **API** opened **large** data **sets** to data **scientists**, but data **inclusion** was **unclear** and twitter **data** is **inherently biased** • the idea of a **digital divide** originally **differentiated** those **who have** access and **those** who **do not** • **Eszter Hargittai** used this **baseline** to consider **second-level digital divides***: those who **have access** but **lack** the **skills** to utilize them* • data **access** is **bound** to those who **hold** the **data** • **data** is often **western-centric** • **scalability** comes at a **cost** • **training** a standard base **model** uses roughly the **same** amount of **energy** as a **transatlantic flight**
Why Scale?	• mass **disruptions** like Covid, can produce **unanticipated growth** • **scalability** is about **creating neutrality** and not all people or systems are the same • **scalability** prioritizes and centers **growth without** needing the system to be **structured** • **scalability** is considered a **morally good** quality of a system • **quantification** is achieved by **identifying** and **manipulating** quantifiable core **elements** or attributes **of** a **system** • **scaled thinking** centers a **capitalistic mindset**

	• **absence** of **scalability** is **equated** with **waste** • **scalability** is **inwoven** with **efficiency**
Data Violence	• the idea of **inclusivity** calls for folding into what **already exists** the **idea** of an **inclusive** system, **without considering if** technology **should exist** at all • **data violence predates** the '**data revolution'** • examples of **data violence** • past **census data** was used to **aid** the **Japanese internment** in the U.S. • **computational tools** used by the **nazis** • **surveillance** systems often **disproportionately surveil certain demographics** more than others • digitally **encoded systems** can **reinforce** these **biases**

Discussion

Where do you see data violence? Take some time and consider what might constitute data violence. Then post about what you realized. What is "violence" and is it possible to mitigate?

I do not know where the tipping point was when the social and cultural climate turned cold on the elderly. Did GenX do this? Did the internet? The open disdain with which the visible public regards senior Americans is a national tragedy that needs immediate attention and remediation.

During the same decades that saw inspiring progress with social inclusion and normalization of the previously socially exiled, the treatment and disparaging rhetoric surrounding our more experienced citizens divulges a group think burdened with distrust and a perception of worthlessness. This attitude manifests in the deplorable ubiquity of locking our predecessors into extended care and group living facilities that feel uncomfortably like jails. Parallel to these attitudes and practices, travels virtually unnoticed, rampant digital marginalization and surveillance leaving a sizable subset of our population isolated and without dignity (Berridge & Grigorovich, 2022).

I want to believe that this is mostly just an awareness issue, and as soon as we can safely and reliably bring care for our aging parents affordably into the home, and a few key influential social figures bring attention to this unfortunate social imbalance, the remediation will self-execute.

I am not influential, socially or otherwise, but maybe I can do some small part to help bring competent and empathy-aware AI-driven and robot-facilitated home healthcare into common use. I believe that artificial intelligence, awareness, and robotic assistance will bring the social attitude toward the elderly in this country back to the place of respect that it deserves, and in doing so bring wisdom and empathy back into our lives and homes.

Berridge, C., & Grigorovich, A. (2022). Algorithmic harms and digital ageism in the use of surveillance technologies in nursing homes. Frontiers in Sociology, 7, 957246. https://doi.org/10.3389/fsoc.2022.957246